

Knowledge Alignment Benchmarking: Evaluating Causal Clinical Reasoning in Multimodal LLMs on Brain MRI

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Abstract

Multimodal large language models (MLLMs) are increasingly evaluated on medical imaging tasks, but almost every existing benchmark scores only final-answer accuracy. Accuracy alone cannot distinguish a model that reasoned correctly from one that pattern-matched the question text or produced reasoning that contradicts its own answer, a distinction that matters in clinical settings where an unfaithful but accidentally correct answer will not generalize to a novel case. We introduce **Knowledge Alignment Benchmarking (KAB)**, a framework for evaluating multimodal clinical reasoning as a causal chain of five phases mirroring how a radiologist works through a brain MRI case: anatomical assessment, lesion localization, diagnostic synthesis, prognostic judgment, and treatment planning. KAB scores models along three axes beyond accuracy: *faithfulness*, measured by grounding a model’s stated reasoning against two authoritative medical ontologies (RadLex and the NCI Thesaurus) and by probing whether a model’s reasoning text alone predicts its own answer; *feedback-adaptation*, measured by injecting a model’s missed clinical concepts and re-querying to see whether it self-corrects; and *soft gating*, which blocks a phase from being scored as valid reasoning if the phases it causally depends on were not themselves answered correctly. In building this framework on the existing OmniBrainBench dataset, we found that its phase groupings do not reflect genuine same-patient continuity (of 6,823 questions in the dataset, only 2 patients contribute a real same-patient multi-phase pair, 4 questions total, both from VQA_RAD and both AIA+LIL only), which undermines any claim about reasoning degrading across a single patient’s causal chain. We describe the construction of a genuine same-patient longitudinal chain dataset built on LUMIERE, a real 91-patient glioblastoma cohort, by drafting multiple-choice items from structured per-patient facts with an LLM and en-

forcing leak control with an automatic stem-leakage auditor. Using this dataset, we run a suite of four checks: a stem-leakage audit, a text-only ablation, an evidence fact-check, and a counterfactual-image substitution, to test directly whether correct answers are grounded in the image rather than in the question text; results from all four checks, on the open-weight model suite, indicate that they are largely not, with the counterfactual check showing that answers which do change under a substituted image mostly fail to track that image’s actual content. This paper presents the KAB framework, the construct-validity problem it surfaces in existing benchmarks, the dataset construction method used to address it, and this image-grounding test suite together with its results to date.

1 Introduction

Multimodal large language models (MLLMs) are being applied to medical imaging tasks (reading a brain MRI, describing a lesion, suggesting a diagnosis) at a pace that has outrun the tools available to evaluate them safely. The dominant evaluation paradigm treats a medical visual question answering (VQA) task exactly like any other multiple-choice benchmark: a model is scored on whether it selects the correct option, and nothing else is measured. This is a critical gap for clinical applications, because final-answer accuracy cannot distinguish between three behaviors that look identical at the level of a selected letter but are clinically distinct:

1. The model correctly perceived the imaging findings and reasoned validly to a diagnosis or plan.
2. The model pattern-matched superficial cues in the question text or answer options, arriving at the right answer without engaging with the image at all.

3. The model guessed, or produced a chain of reasoning that is internally inconsistent with the answer it ultimately gave.

Behaviors (2) and (3) are well documented outside the medical domain: crowd-sourced natural language inference datasets contain annotation artifacts that let models reach high accuracy from the hypothesis sentence alone (Gururangan et al., 2018), and models trained on such data adopt syntactic heuristics that collapse under adversarial evaluation (McCoy et al., 2019). Chain-of-thought reasoning inherits the same failure mode: a model’s stated reasoning frequently does not reflect the computation that actually produced its answer (Lanham et al., 2023); this occurs on naturally worded prompts without any adversarial construction (Arcuschin et al., 2025), and intermediate reasoning structures behave as influential context rather than as stable causal mediators of the final decision (Somov et al., 2026). In the medical multimodal setting specifically, cascading errors from early-stage reasoning failures have been identified as a leading cause of incorrect predictions (Jung et al., 2026), and models have been shown to rely on textual or visual shortcuts rather than the pathological regions a question actually asks about (Nguyen et al., 2025). In a clinical setting, an accurate-but-unfaithful answer is not merely a benchmark artifact; it is a model that cannot be trusted on the next case that differs even slightly from the ones it was evaluated on.

We argue that a medical MLLM benchmark should evaluate reasoning the way a clinician’s diagnostic process is actually structured: as a sequential, error-compounding chain, where later judgments are only valid if earlier ones were correct (Croskerry, 2009). We introduce **Knowledge Alignment Benchmarking (KAB)**, a framework that decomposes a brain MRI case into five clinical phases: **Anatomical and Imaging Assessment (AIA)**, **Lesion Identification and Localization (LIL)**, **Diagnostic Synthesis and Causal Reasoning (DSCR)**, **Prognostic Judgment and Risk Forecasting (PJRF)**, and **Therapeutic Cycle Management (TCM)**. It scores a model on three dimensions beyond accuracy: whether its reasoning is grounded in real clinical vocabulary and is self-consistent (**faithfulness**), whether it can incorporate a targeted clinical correction when it fails (**feedback-adaptation**), and whether its chain of answers is causally coherent rather than a set of

independently-guessed phases (**soft gating**). Figure 1 summarizes the five-phase chain and how each of the three metrics attaches to it.

In the course of building KAB on top of the existing OmniBrainBench dataset (Peng et al., 2026), we discovered a construct-validity problem that we believe generalizes beyond our own project: OmniBrainBench’s phase-to-case grouping does not track same-patient identity. Of 6,823 questions spanning 15 source sub-corpora, only 2 patients (4 questions) form a genuine same-patient pair across two different clinical phases, and zero span three or more phases. What the current grouping evaluates as a chain is stitched together from different real patients’ questions within the same origin sub-corpus, so a model gated out of a later phase after failing an earlier one is demonstrating an artifact of this aggregation, not evidence of causal reasoning failure for one patient. We report this problem in detail (Section 4) and describe our resolution: a genuine same-patient longitudinal chain dataset built on LUMIERE (Suter et al., 2022), a 91-patient real-world glioblastoma cohort with longitudinal imaging, expert-assigned RANO treatment-response ratings, survival data, and Stupp-protocol treatment records, using an LLM-drafting construction pipeline with an automatic leak-control audit (Section 5).

This paper makes four contributions:

- We identify and quantify a construct-validity problem in an existing brain-MRI MLLM benchmark that is likely to recur in any dataset assembled from single-task source corpora.
- We present a reusable construction method (automatic fact extraction, LLM item drafting, and an automatic leak-control audit) for turning a raw longitudinal clinical imaging dataset into a construct-valid multi-phase reasoning benchmark.
- We present the KAB framework itself: ontology-grounded faithfulness scoring against RadLex and the NCI Thesaurus, a feedback-adaptation loop that distinguishes a knowledge gap from an activation failure, and soft gating that operationalizes the causal dependency between clinical phases.
- Once chains are genuine, KAB’s degradation and adaptation measurements become answers to a well-posed question: whether a model’s reasoning at a later phase actually follows from what

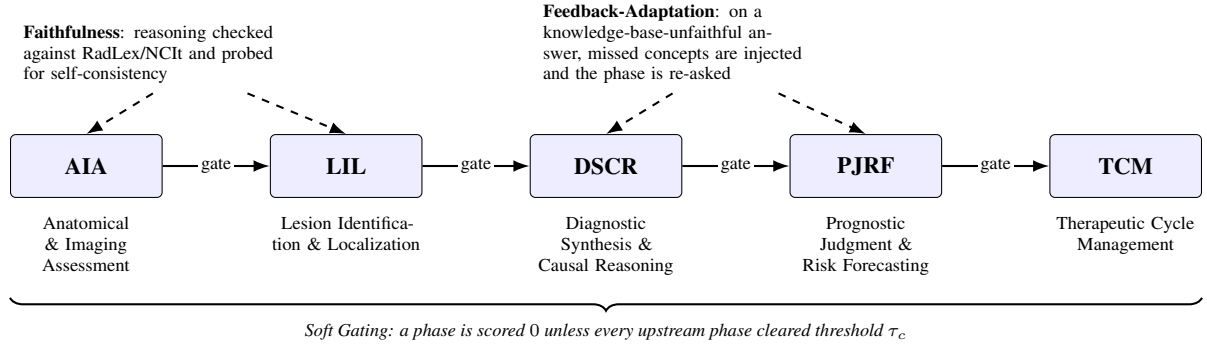


Figure 1: The KAB framework: a brain-MRI case is evaluated as a five-phase causal chain (bottom), with three metrics attached to it (top and below): faithfulness (grounding + self-consistency), feedback-adaptation (a targeted re-query on failure), and soft gating (a phase counts only if its upstream dependencies were answered correctly).

it established earlier for the same patient, rather than an artifact of how a dataset happens to be grouped.

Evaluation across the full model suite (Section 6) remains incomplete (Section 8); this paper reports results from the open-weight subset.

2 Related Work

Shortcut learning and reasoning artifacts. The observation that a model can reach a correct label without engaging with the intended reasoning task predates large language models. Gururangan et al. (2018) showed that natural language inference hypotheses alone, without the premise, carry enough annotation artifacts to substantially outperform chance. McCoy et al. (2019) showed that models trained on such data adopt syntactic heuristics (lexical overlap, subsequence, constituent) that fail systematically on their HANS diagnostic set. KAB’s answer-recoverability probe and correct-but-unrecoverable flag are a direct descendant of this line of work, applied to a setting where the *hypothesis* is a model’s own free-text reasoning rather than a dataset artifact.

Chain-of-thought faithfulness. Lanham et al. (2023) introduced systematic tests for whether a model’s chain-of-thought causally affects its final answer, finding that stated reasoning frequently does not reflect what actually drove the answer. Arcuschin et al. (2025) extended this to naturally-worded, non-adversarial prompts, reporting unfaithfulness rates up to 13% in production models. Somov et al. (2026) evaluated schema-guided pipelines under causal intervention on intermediate structures and found models fail to update their final decision after the intervention in up

to 60% of cases, concluding that intermediate reasoning functions as influential context rather than a stable causal mediator. KAB’s answer-recoverability probe is methodologically closest to this line: rather than intervening on the reasoning text, we withhold everything except the reasoning text and ask whether it alone reproduces the original answer.

Cascading failure in medical multimodal reasoning. Two recent works are close enough to KAB’s gating mechanism to warrant explicit comparison. Jung et al. (2026) propose Medical Reasoning-aware Policy Optimization (MRPO), a reinforcement-learning method that assigns exponentially larger penalties to earlier invalid reasoning steps during training, directly targeting the same cascading-failure phenomenon that motivates KAB’s soft gating. Nguyen et al. (2025) introduce perturbation tests that detect textual and visual shortcut learning in medical VQA models. Both works target the same underlying phenomenon, early-stage reasoning failure propagating into later errors, but neither addresses *dataset construct validity*: whether the multi-step cases used for training or evaluation actually represent one patient’s causal trajectory in the first place. MRPO is a training-time intervention that presupposes its step-labeled data is a genuine causal chain; our contribution is upstream of that assumption. KAB additionally couples cascading-failure detection to ontology-grounded faithfulness scoring and a feedback-adaptation loop, neither of which is part of either prior work.

Clinical cognition as motivation for phase structure. The five-phase decomposition in KAB is not an arbitrary engineering choice. Croskerry

(2009) describes clinical diagnostic reasoning under a dual-process model in which sequential, error-compounding cognitive steps, rather than a single holistic judgment, characterize real diagnostic error. This gives a clinical-science grounding for treating AIA → LIL → DSCR → PJRF → TCM as a genuine dependency chain rather than a convenient dataset partition.

Brain MRI benchmarks and same-patient continuity. OmniBrainBench (Peng et al., 2026) is, to our knowledge, the largest closed-ended brain-MRI MLLM benchmark organized around a multi-phase clinical workflow, and is the dataset on which we build the KAB framework. NeuroQA (Abbasi et al., 2026) is substantially larger (56,953 QA pairs, 12,977 subjects) and includes a *Longitudinal* question category, but this category compares two timepoints rather than a full causal chain, and its authors explicitly state the benchmark is not intended to support claims about treatment planning. UCSF-PDGM-VQA (Ghosh et al., 2026) provides 2,387 QA pairs over 473 glioma studies but at a single timepoint per patient, with no prognostic or treatment layer. LUMIERE (Suter et al., 2022) and MU-Glioma-Post (Washington University School of Medicine et al., 2025) both provide genuine longitudinal, real-patient continuity with treatment and outcome data, but neither ships a question-answering layer. We are not aware of a published dataset that combines genuine same-patient continuity across imaging, diagnosis, prognosis, and treatment with a ready-made multiple-choice question layer; Section 5 describes our construction of one on top of LUMIERE.

3 The KAB Framework

KAB evaluates a model on a per-case basis, where a case is one patient’s progression through some subset of the five clinical phases. For each question in a case, a model receives the relevant image(s), the question, the answer options, and a structured summary of its own answers from upstream phases in the same case (the *chain context*). It must return a structured response containing an answer, a description of what it observed in the image (visual_grounding), and free-text reasoning connecting the observation to the answer. On top of this base interaction, KAB computes three families of metrics.

3.1 Faithfulness

Answer recoverability. After a model answers, we issue a second call, the recoverability probe, containing *only* the model’s own reasoning text (no image, no question, no answer options) and ask which answer that reasoning supports. If the probe’s predicted answer matches the model’s original answer, the response is *recoverable*: the stated reasoning is sufficient, on its own, for a downstream reader (here, the probe model) to reconstruct the same conclusion. A mismatch means the model’s justification is disconnected from the answer it actually gave. This establishes textual self-consistency between an answer and its stated reasoning; it does not establish that the reasoning caused the answer, since a response can restate its own conclusion in different words without that restatement having driven the decision. We separately flag *correct-but-unrecoverable* responses, a right answer paired with reasoning that does not support it, as the most clinically dangerous category, since such a response offers no evidence it will generalize to a case that differs even slightly from the one it was scored on.

Ontology term coverage. This is KAB’s central grounding-adjacent metric; despite the name, it measures term *validity*, not correctness (see Section 8). Every diagnostic phase has an authoritative source of real clinical vocabulary: RadLex (Lan-glotz, 2006) for radiology anatomy, imaging modalities, and lesion descriptors (routed to the AIA and LIL phases), and the NCI Thesaurus (NCIt) for tumor classification, WHO grading, molecular markers, staging, and treatment (routed to DSCR, PJRF, and TCM). NCIt was chosen over UMLS because OmniBrainBench and LUMIERE are both predominantly glioma/glioblastoma datasets, for which NCIt has the deepest freely-available coverage, and because it requires no account or API key, removing a reproducibility barrier. We concatenate a model’s visual_grounding and reasoning fields, tokenize the text (preserving hyphenated compounds such as *ring-enhancing* as single tokens), and run a greedy longest-match scan against the union of both knowledge bases, trying 4-grams down to 1-grams with no reuse of consumed token positions. This yields extracted_concepts: every KB term the model used regardless of phase. We then take matched_concepts as the subset of extracted_concepts present in the phase-

appropriate KB, and compute

$$\text{OntCoverage} = \frac{|\text{matched_concepts}|}{|\text{extracted_concepts}|}. \quad (1)$$

A low score indicates the model is not using vocabulary appropriate to the current diagnostic phase, for example still describing anatomy at the diagnosis phase rather than committing to tumor-classification language; unrecognized assertions are excluded from the denominator entirely, so this score says nothing about whether the model’s *unrecognized* content is accurate, only about the recognized fraction.

Answer-term overlap. Ontology term coverage measures whether a model used *real* clinical terms; answer-term overlap measures whether it used ones that also appear in *this case’s correct answer text* — a proxy for on-target vocabulary, not for whether the surrounding explanation is a genuinely correct account of the case. Of the matched phase-KB concepts, we compute the fraction that also appear in the correct answer text:

$$\text{AnswerTermOverlap} = \frac{|\text{on-target matches}|}{|\text{matched_concepts}|}. \quad (2)$$

A model can score high on ontology term coverage while scoring low on answer-term overlap: using plausible-sounding real clinical language that is nonetheless wrong for the case at hand. The converse is also possible: repeating answer-option wording verbatim scores high on this metric without necessarily reflecting reasoning.

Upstream-term repetition. For each phase after AIA, we check what fraction of the key terms from the correct answers of *upstream* phases are referenced in the current phase’s reasoning text. This measures textual repetition of upstream terms, not demonstrated reasoning dependency: a model that identified a lesion in the right temporal lobe during LIL but never references that location during DSCR is *at least* answering each phase in isolation rather than visibly carrying context forward, but the converse does not hold, repeating an upstream term does not by itself show the current answer was reasoned *from* it, nor does omitting one show independent reasoning.

3.2 Feedback-Based Adaptation

The metrics above establish whether a model’s stated reasoning uses valid, on-target clinical vocabulary and is self-consistent with its answer, not

whether the model reasoned correctly; the adaptation loop asks a different question: whether a model with low ontology term coverage can be corrected when the missing vocabulary is handed to it directly. The loop triggers when a response has low ontology term coverage (OntCoverage below a threshold τ_g , default 0.3) *and* has a non-empty set of missed concepts, defined as phase-KB terms present in the correct answer but absent from the model’s output. When triggered, we re-issue the original prompt augmented with the model’s prior answer and reasoning and an explicit CLINICAL KNOWLEDGE CORRECTION block listing the missed concepts, and ask the model to re-examine the image in light of this correction. We record whether the model changed its answer (adaptation_changed) and whether the revised answer is correct (adaptation_correct), and report the *adaptation rate*: the fraction of triggered questions that the model corrects to the right answer. We describe this as performance under a targeted, answer-derived hint; correcting after such a hint does not by itself establish that the model already possessed the relevant knowledge and merely failed to activate it, since the hint also supplies new information. Distinguishing a genuine *knowledge gap* from an *activation failure* would require comparing this targeted-hint condition against a generic-retry control (re-asking without new information) and an irrelevant-hint control (a correction block naming concepts not relevant to the case); we have not yet run these controls, so we report the targeted-hint adaptation rate alone rather than the knowledge-gap/activation-failure distinction.

3.3 Soft Gating

Because the five phases form a causal chain, scoring a diagnosis (DSCR) as valid reasoning after a model has already failed to identify the correct anatomy (AIA) produces a misleading result: the diagnosis was not built on correct upstream perception. Soft gating enforces this directly: a phase is blocked (gated_out, scored 0.0) unless every phase it depends on scored at or above a gating threshold τ_c (default 0.5). AIA has no upstream dependency and always runs; LIL requires AIA; DSCR requires AIA and LIL; PJRF requires AIA, LIL, and DSCR; TCM requires all four. *Chain completion rate*, reported at the case level, is the fraction of cases in which every phase ran and cleared its gate; a model with a high chain completion rate is demonstrating coherent end-to-end reasoning

rather than isolated per-phase competence. Gating can be disabled for a flat accuracy baseline.

4 Chain Validity in OmniBrainBench

The KAB framework as described in Section 3 presupposes that a case’s five phases represent one patient’s actual clinical trajectory: a model that fails AIA for a given patient should be gated out of that same patient’s DSCR question, not an unrelated patient’s. We investigated whether OmniBrainBench (Peng et al., 2026) supports this premise directly, by checking whether any image is shared across questions tagged with different clinical phases within a grouped case, the only way genuine same-patient, multi-phase continuity could hold.

OmniBrainBench ships 9,527 total QA pairs; we use its 6,823-question *closed-ended* (multiple-choice) subset, since KAB’s gating and scoring require answer options, and exclude the remaining 2,704 open-ended pairs. Of these 6,823 questions, drawn from 15 source sub-corpora (including PubMedVision, BraTS2021, RSNA, ISLES2022, and VQA_RAD, among others), we found only 2 patients whose images genuinely appear in questions from two different phases, both from VQA_RAD, each contributing one AIA question and one LIL question (4 questions total across the 2 pairs). We repeated the analysis with a second, independent method (extracting real per-patient case identifiers by stripping known modality/view/slice suffixes from filenames on a per-corpus basis) and obtained the same result: 2 genuine multi-phase same-patient cases in the entire dataset (4 questions), zero spanning three or more phases, zero spanning all five. The root cause is structural: each of the 15 source sub-corpora was constructed for one specific clinical task (e.g., BraTS for tumor characterization, RSNA for hemorrhage classification, ISLES for stroke), so a given real patient within a sub-corpus is essentially only ever asked one phase-type of question. `source_file` is a field present in OmniBrainBench’s own released data, denoting each question’s origin annotation batch; using it to represent same-patient identity for a multi-phase causal chain is our own KAB-side interpretation, not a claim OmniBrainBench itself makes about that field. Under that interpretation, `source_file` in practice groups together an entire origin sub-corpus, standing in for a single patient; what this grouping evaluates as a chain is stitched from *different real patients’* questions, not one patient’s actual

diagnostic journey. A model gated out of a later phase after failing an earlier one under this grouping demonstrates nothing about causal reasoning degradation for any single patient.

We searched for a replacement or supplementary dataset with genuine same-patient, multi-phase continuity and a ready-made question layer (Section 2); none exists. This motivates the construction described next.

5 Constructing a Genuine Causal-Chain Dataset from LUMIERE

We build a genuine same-patient longitudinal chain dataset on LUMIERE (Suter et al., 2022), a single-center dataset of 91 real glioblastoma patients with 638 study dates, automated multi-sequence tumor segmentation, and clinician-assigned RANO treatment-response ratings with documented rationale. Figure 2 summarizes the automatic construction pipeline described in this section.

Fact extraction. For each patient, we extract per-phase facts directly from LUMIERE’s shipped data with full provenance, requiring no additional annotation for the perception-oriented phases: LUMIERE’s tumor segmentation masks are already registered to a standard MNI152 atlas space (the same space used by standard neuroanatomical atlases), so lesion-location facts reduce to a voxel-index lookup rather than a registration step, and per-timepoint tumor volumes ship precomputed. DSCR facts use LUMIERE’s existing expert-assigned RANO ratings directly rather than requiring new diagnostic annotation. PJRF and TCM facts draw on the dataset’s structured survival and Stupp-protocol treatment records.

Cohort. Of LUMIERE’s 91 patients, 55 satisfy our patient-level eligibility criterion: at least 3 RANO response-rated follow-up timepoints (for a genuine multi-timepoint chain) and at least one timepoint with complete multi-sequence imaging, checked independently rather than requiring both at the same timepoint; we draft from this full 55-patient eligible cohort rather than a smaller convenience sample to maximize the size of the resulting item set. This patient-level criterion does not guarantee that a single timepoint carries *both* a RANO rating and complete imaging, which fact extraction additionally requires; one patient (Patient-025) has no timepoint satisfying both conditions simultaneously and is dropped at extraction time, leaving 54

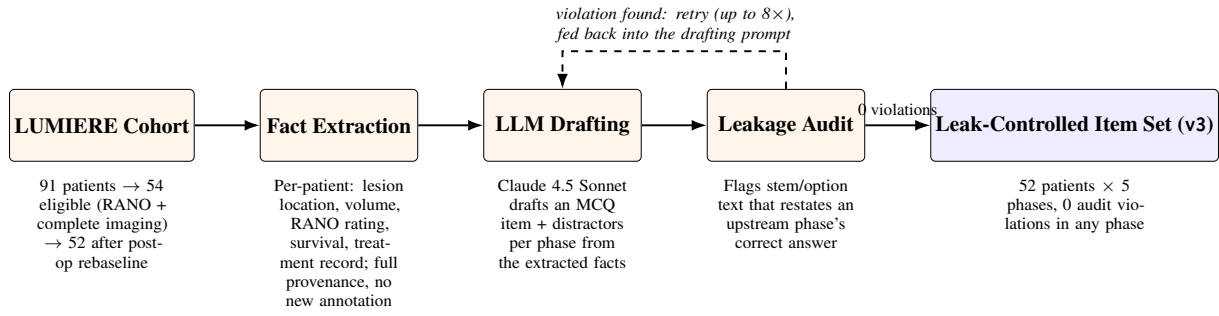


Figure 2: Construction pipeline for the leak-controlled (v3) LUMIERE item set. Structured per-patient facts are extracted from LUMIERE, an LLM drafts multiple-choice items from those facts, and an automatic auditor flags any item that restates an upstream phase’s correct answer, feeding violations back into the drafting prompt for up to eight retries.

patients and 269 drafted multiple-choice items in the item set described below.

Timepoint sampling. We initially selected each patient’s most recent RANO-rated timepoint, but found this made nearly every DSCR answer *progressive disease*, since glioblastoma patients overwhelmingly trend toward progression by their final scan, a sampling choice that would have made the DSCR phase trivially guessable regardless of a model’s actual reasoning. We instead sample a random eligible response-rated timepoint per patient (fixed seed), which yields a substantially more balanced label distribution across the 54-patient cohort: 37 progressive disease, 9 stable disease, 4 partial response, and 4 complete response. We additionally found and fixed a related bug in which a randomly sampled RANO-rated timepoint could lack matching complete imaging, silently falling back to imaging from a *different* date than the one the RANO rating and DSCR facts describe; we now constrain sampling to timepoints with both a rating and complete imaging, and verified zero such mismatches across the full extracted cohort.

LLM drafting. A large language model (Claude 4.5 Sonnet, accessed via a local LiteLLM proxy) drafts multiple-choice items per phase from the extracted per-patient facts, including plausible distractors. The perception-adjacent phases (AIA, LIL) and the diagnosis phase (DSCR) are close to programmatically derivable from LUMIERE’s shipped segmentation and RANO data. The prognosis (PJRF) and treatment (TCM) phases instead require distractors informed by the glioblastoma prognostic and treatment literature (particularly around confounds such as pseudoprogression, a well-known source of misjudgment even for prac-

ticing radiologists), which the drafting prompt encodes explicitly for these two phases.

Stem-leakage audit and a leak-controlled item set. Manual inspection of the drafted items surfaced a systematic construction flaw: the drafter wrote each phase’s question stem as a self-contained clinical vignette, which led it to restate findings established in earlier phases of the same chain. We built an automatic auditor that flags, per phase, any stem or option text containing content it should not have access to: a numeric percent volume change, a raw volume or region name, a change-direction word, a RANO label, or a stated survival duration, checked both against the phase’s own correct answer (*self-leakage*) and against the correct answers of upstream phases in the same chain (*upstream leakage*). Applying this auditor to the initial item set (v2) found leakage in a majority of DSCR, PJRF, and TCM items and a smaller fraction of LIL items; a lexical baseline that guesses the option with the highest word overlap with the stem performs at or below chance on the same items, confirming the leak is semantic (e.g., a stated percent volume increase implying *progressive disease*) rather than a copy-paste artifact a naive overlap check would already catch.

We traced the cause to the drafting prompt itself, which supplied each phase’s drafting call with the correct answers of upstream phases (necessary for constructing a coherent chain) without instructing the model not to restate them, and, for DSCR specifically, explicitly asked the model to let the reader *infer* the response label from volume-change facts it had just been shown. We built a revised, leak-controlled item set (v3) that keeps the same upstream-truth context available to the drafter but forbids restating it in the visible stem or options, re-

drafting automatically (up to eight attempts) whenever the auditor flags a violation and feeding the specific violated rule back into the retry prompt. The DSCR stem, which the auditor showed was hardest to keep leak-free under free-form drafting, is instead built from a fixed template populated only from structured extent-of-resection, re-resection, and radiotherapy-timing fields, with no correct-answer information available to leak. The leak-controlled item set clears the auditor with zero violations across every phase.

Image rendering. For each timepoint shown to a model, we render a single 2D axial slice: the slice with the largest tumor-mask cross-sectional area (rather than a fixed anatomical level), taken from the atlas-registered, contrast-enhanced T1 (CT1) sequence, rotated 90° to standard radiological display orientation, and intensity-normalized by clipping to the 1st-99th percentile of the slice’s own voxel values before scaling to 8-bit grayscale. Each patient contributes exactly one such slice per timepoint (baseline and follow-up), saved as a PNG; no window/level preset beyond this percentile clip is applied, and no multi-slice or 3D rendering is used.

Post-operative rebaselining. Building the leak-controlled item set surfaced a second issue upstream of leakage: the LIL phase’s volume-change baseline was, for 42 of 54 patients, a pre-operative scan, so the measured volume change mostly captured the surgical resection rather than the treatment response that the RANO rating (DSCR) is actually judged against. We recompute the LIL baseline as each patient’s latest imaged post-operative scan prior to the follow-up (advancing past any re-resection that occurs before the follow-up date), changing the baseline for 44 of 54 patients and shifting the cohort’s volume-change direction from a resection-dominated skew to a more balanced 20 patients increasing versus 32 decreasing. Two patients (Patient-049, Patient-068) lack a post-operative scan with matching segmentation and are excluded from the leak-controlled item set rather than mixed in on a pre-operative baseline, leaving 52 patients. Downstream DSCR, PJRF, and TCM facts and answer keys, which do not depend on the LIL baseline, are unchanged by this step.

Display-orientation note. Rendered slices follow radiological convention (patient’s right shown on the image’s left), which the original prompt never stated explicitly. For the leak-controlled item

set only, we prepend a one-line statement of this convention to the image content sent to the model, leaving OmniBrainBench and the initial LUMIERE item set unaffected.

Contribution. Once constructed, this is, to our knowledge, the first dataset combining genuine same-patient continuity with a full imaging-to-treatment question layer, closing a gap that persists across every existing candidate we are aware of (Section 2). Beyond the dataset itself, we position the LLM-drafting-plus-automatic-audit pipeline as a reusable construction method: it is a substantially cheaper way to turn a raw longitudinal clinical dataset into a construct-valid multi-phase reasoning benchmark than a fully manual annotation effort.

6 Experimental Setup

Models. We evaluate a suite spanning two categories: three proprietary models accessed via API or a LiteLLM proxy (GPT-5, Claude 4.5 Sonnet, Gemini 2.5 Pro), and four open-weight models served through Ollama at 4-bit quantization (MedGemma-4B, the suite’s only medical-domain fine-tune, and three general-purpose models: Gemma-3-12B, Gemma-3-27B, and Llama-4-Scout, a 109B-parameter mixture-of-experts model with 17B active parameters). The exact served tags are medgemma:4b, gemma3:12b, gemma3:27b, and llama4:scout (Ollama’s official builds, default Q4_K_M quantization). All models are accessed through the same OpenAI-compatible interface so the evaluation pipeline is identical regardless of backend. Decoding is greedy (temperature 0) with a 800-token response budget, and the response schema requests answer before visual_grounding/reasoning so that a truncated response still contains a complete answer field.

Infrastructure. Open-weight model serving and inference runs on single-GPU SLURM jobs (NVIDIA RTX PRO 6000); every model in the open-weight suite serves at 4-bit quantization on a single GPU, so multi-GPU tensor parallelism is not required.

Failed-response handling. An API call is retried up to 3 times with exponential backoff (5s/10s/20s) before being treated as empty. If a response’s JSON fails to parse (observed causes: MAX_TOKENS truncation mid-field, and un-escaped literal newlines inside a string value, the latter recovered losslessly

by a lenient non-strict parse), we attempt to salvage the answer letter alone via a regex on the "answer" field, which the schema places first; if that also fails, the item is scored incorrect rather than excluded, so a model prone to malformed output is not given a smaller, easier denominator.

Knowledge bases. RadLex (RSNA, English-filtered OWL release, downloaded 2026-08-05; the release file carries no embedded owl:versionInfo, so we cite it by download date) and the NCI Thesaurus (NIH NCI, OWL release, version 26.07d) are used as-is, with no stub or fallback vocabulary: the KB aligner requires both files to be present and raises an error rather than silently degrading if either is missing.

7 Testing Whether Correct Answers Are Image-Grounded

Faithfulness (Section 3.1) tests whether a model's stated reasoning is internally consistent with its answer; it does not test whether that reasoning is actually derived from the image. The answer-recoverability probe in particular is a weak instrument for this specific question by construction: it asks the same model to predict its own answer from its own reasoning text, so any response whose reasoning happens to restate the chosen option passes the probe regardless of whether an image was ever consulted, independent of how often that occurs in practice. We therefore built a separate suite of four checks, run on both the initial and leak-controlled LUMIERE item sets (Section 5) and applied identically across the full model suite (Section 6), to test image-grounding directly rather than inferring it from a self-consistency probe.

Stem-leakage audit. Described in Section 5 as the mechanism used to build the leak-controlled item set, the same auditor doubles as a grounding diagnostic: an item whose stem or upstream chain context already states the correct answer in words is not testing image use at all, regardless of what a model's accuracy or faithfulness score reports. Running the two item sets through this single, unmodified auditor therefore also serves as a controlled before/after manipulation for the three checks below.

Text-only ablation. We re-run the evaluation pipeline with all images stripped before the main call (question stem, answer options, and chain

context from upstream phases are otherwise unchanged), and compare accuracy against the matched image-present run on the same items. Both arms run with gating disabled, so every phase is scored for every patient regardless of upstream correctness; gating a phase's score would not change what a model is shown or answers, only whether that answer counts, and disabling it here avoids shrinking the comparison sample on later phases. The comparison isolates a specific claim: if the patient-clustered paired difference between text-only and image-present accuracy falls within a pre-specified equivalence margin, the image is not necessary for the model to reach its answer on this item set, whatever its stated reasoning claims; a non-significant difference alone would not support this claim, since failing to detect a difference is not the same as establishing equivalence.

Evidence fact-check. Necessity and use are different claims: an image could be unnecessary on average while still being consulted, so we separately audit the content of a model's stated evidence against the same structured per-patient facts used to draft the items (Section 5). For each factual claim in a correct answer's visual_grounding and reasoning fields (hemisphere/region statements and any measurement with units), we assign one of four labels: *copied*, the claim's content already appears in the stem, answer options, or the upstream chain context the model was shown; *independently supported*, the claim was not shown to the model but matches the ground-truth facts; *fabricated*, the claim matches no ground-truth fact; or *laterality flip*, a hemisphere claim that disagrees with the segmentation under the assumption that image left corresponds to patient right. We score the fourth category separately from ordinary fabrication because the rendered slices follow radiological display convention without stating so in the prompt for the initial item set (fixed for the leak-controlled set, Section 5): a hemisphere claim that is wrong under one convention and correct under the other reflects a display-convention ambiguity in the stimulus, not a perceptual error, and conflating the two would misattribute a prompt-design gap to the model. RANO-criteria thresholds quoted verbatim from clinical guidelines (e.g., " $\geq 25\%$ increase") are excluded from the fabricated-measurement count, since they restate a rule rather than assert a patient-specific fact. The audit itself is deterministic pattern matching, not an LLM judge:

numeric claims are extracted and compared against the structured ground-truth facts and the shown-text pool by value, and hemisphere claims are checked against the segmentation’s own left/right extent; this makes the audit’s category assignment fully reproducible from a response’s text, but also means it does not attempt to interpret paraphrase or synonymy beyond this direct matching, a source of misclassification risk we return to in Section 8.

Counterfactual-image substitution. The strongest of the four checks: rather than removing the image (ablation) or auditing what a model says about it (fact-check), we substitute a *different, real patient’s* own image in place of the original patient’s, while leaving every text field (stem, answer options, chain context, and the image labels naming which timepoint is baseline versus follow-up) exactly as the original patient’s. A model that is actually using the image should be more likely to answer consistently with what is now in front of it; a model relying on the (unchanged) text should not. We pair every patient in the leak-controlled item set with a partner via a deterministic, seeded assignment: because the item set’s post-operative rebaselining (Section 5) yields a near-even split of patients whose lesion volume increases versus decreases between baseline and follow-up, every patient can be assigned a partner with the *opposite* volume-change direction, guaranteeing a substantively different image is shown in every case; we additionally report, without optimizing for it, how often the pairing also yields a different DSCR response-assessment label. The substitution is applied to all five phases for completeness, using each partner’s own rendered timepoint images for the phase in question, and evaluated non-gated for the same sample-preservation reason given above. Our primary comparison is the LIL and DSCR phases, the two phases whose correct answer is most directly a function of image content; AIA’s correct answer does not vary with the image shown (a single fixed sequence-protocol string across the cohort) and is included only for completeness, and PJRF/TCM’s own-image effect is confounded by upstream phases that are simultaneously evaluated under their own swapped image, so a chain-context effect from an upstream swap cannot be fully separated from a same-phase effect there.

Our primary outcome is the *flip rate*: the fraction of items on which a model’s answer under

the substituted image differs from its answer under the patient’s own image. We additionally check, among items that do flip, whether the new answer matches what the substituted patient’s own facts would actually support, which distinguishes a flip driven by genuine image content from one driven by decoding variance alone. We pre-registered the joint reading of all four checks before running any of them: text-only accuracy equivalent to image-present accuracy within a pre-specified margin (we use $\pm 8\text{pp}$ on the patient-clustered paired difference, Section 8), evidence classified mostly as copied rather than independently supported, and a low, non-truth-tracking counterfactual flip rate together would indicate that correct answers on this item set are not meaningfully image-grounded; a clear text-only accuracy drop together with independently-supported evidence and a counterfactual flip rate that tracks the substituted patient’s truth would instead indicate genuine grounding, in which case we would not report a non-grounding claim regardless of what the first three checks alone suggested.

8 Results

We report results from all four grounding-test checks described in Section 7 (the stem-leakage audit, the text-only ablation, the evidence fact-check, and the counterfactual-image substitution) together with the KAB framework’s own faithfulness, alignment, and adaptation metrics (Section 8, “KAB framework metrics” below). All numbers below use the leak-controlled (v3) item set unless stated otherwise, so that a leftover leakage effect cannot be mistaken for a grounding effect. These results cover the open-weight model suite only; the proprietary models (Section 6) have not yet been run (Section 8).

Stem-leakage audit. The initial item set (v2) failed the auditor at rates too high to treat its accuracy numbers as informative about image use: 0% of AIA items leaked, but 13% (7/54) of LIL items self-leaked, 98% of DSCR items restated an upstream phase’s exact percent volume change, 85% of PJRF items either stated the patient’s actual survival duration directly (31%) or named the upstream RANO label (72%), and 98% of TCM items restated LIL’s volume-change percentage or RANO label. The leak-controlled item set (v3), built with the same auditor folded into the drafting retry loop (Section 5), cleared every phase with zero violations. All results below use v3.

Model	Image	Text-only	Δ
Llama-4-Scout	49%	52%	+3
Gemma-3-27B	54%	50%	-3
Gemma-3-12B	48%	51%	+3
MedGemma-4B	33%	36%	+3

Table 1: Non-gated overall accuracy on the leak-controlled (v3) item set, with images present versus text-only. Δ = text-only – image, in percentage points.

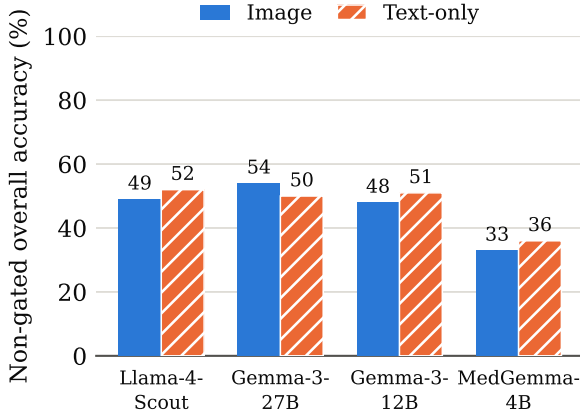


Figure 3: The same comparison as Table 1, drawn on the full 0-100% scale so the size of the gap between conditions is not visually exaggerated: for every model, the image and text-only bars are within a few points of each other.

Text-only ablation. Table 1 and Figure 3 compare non-gated overall accuracy on v3 with and without images, for all four open-weight models. Three of four models score at least as high without the image as with it (Llama-4-Scout, Gemma-3-12B, and MedGemma-4B each +3pp); Gemma-3-27B is the only model with an image advantage (-3pp text-only), driven mainly by AIA (86% image versus 77% text-only), and even that gap falls inside the 95% Wilson confidence interval at $n = 52$. No model shows an image advantage outside its own confidence interval on any phase. Because each patient contributes one question per phase, overall (pooled across phases) accuracy comparisons treat 5 correlated observations per patient as independent unless clustering is preserved; we therefore also compute the image-minus-text-only difference *per patient* (mean accuracy across that patient’s 5 phase-questions) and derive a patient-clustered 95% CI via bootstrap resampling of patients (10,000 resamples): MedGemma-4B -1.9pp [-5.8, +1.9], Gemma-3-12B -2.7pp [-7.7, +2.3], Gemma-3-27B +3.1pp [-0.8, +7.3], Llama-4-Scout -3.1pp

[-7.3, +1.2]. None of these intervals exclude a difference of up to about 7-8 percentage points in either direction; pre-specifying an equivalence margin of ± 8 pp, all four models’ intervals fall within it, but none would at a tighter ± 5 pp margin. We report both the interval and this explicit margin dependency rather than asserting equivalence from the interval alone, since $n = 52$ patients gives limited power to rule out a moderate true effect. At the level of individual questions rather than aggregate accuracy, the pattern is sharper still: the same question receives an identical answer with and without the image for 87-91% of items, for every model. Among the minority of items where removing the image changes the answer, *right only with image* and *right only without image* occur at comparable rates (55 versus 67, summed across models), the pattern expected if the image swap behaves like decoding noise rather than the removal of information the model actually needed, though we have not yet run repeated trials under unchanged inputs to confirm decoding noise is in fact the source of these swaps rather than genuine sensitivity to the image (Section 8).

For context, v3 items are substantially harder than v2 for every model (image-present overall accuracy down 13-19 percentage points, e.g., Llama-4-Scout 67% $\rightarrow$ 49%, MedGemma-4B 45% $\rightarrow$ 33%), consistent with the stem-leakage fix removing an exploitable shortcut rather than removing task difficulty. This drop is a validity correction to the earlier item set, not itself evidence about image use; we report it only to establish that the narrow with/without-image gap on v3 is not an artifact of v3 being an easier, less discriminating test.

Evidence fact-check. Restricting to correct answers on v3, the fraction of stated visual_grounding/reasoning claims classified copied (already present in the stem, answer options, or chain context the model was shown) ranges from 43% to 90% across the four models. The complementary category that would demonstrate genuine image use, independently supported, accounts for only 0-8% of claims. Gemma-3-27B is an outlier within this pattern: rather than copying, it more often states no checkable claim at all, which is consistent with, not contrary to, the same underlying picture: its stated evidence rarely contains a fact it could only have obtained from the image either way.

Counterfactual-image substitution. For each patient, we substitute a different real patient’s own image (the pairing guarantees the opposite lesion-volume-change direction, Section 7) while leaving the stem, answer options, and chain context unchanged, and measure the *flip rate*: how often the answer changes. AIA’s correct answer does not depend on the image at all (a fixed sequence-protocol string across the cohort), so its 10-17% flip rate across models is the rate attributable to reformatting/decoding variance alone, not to image content. LIL (12-42%) and DSCR (6-27%) flip more often than this baseline for three of four models, indicating the substituted image is not simply being ignored. The decisive question is whether those flips track the substituted patient’s actual truth: among flipped LIL and DSCR items, we check whether the new answer’s volume-change direction or RANO label matches what the substituted patient’s own facts support. Pooled across all four models and both phases, the truth-tracking rate is 35.6% (26/73, 95% Wilson CI [25.6, 47.1]) — an interval that sits entirely below the 50% rate expected by chance on this binary-ish distinction, not above it. Per model, only Llama-4-Scout reaches chance (50% on both LIL and DSCR, $n = 6$ each, a small sample); every other model is below 50% on both phases (MedGemma-4B 25%/0%, Gemma-3-12B 38%/36%, Gemma-3-27B 21%/46%, LIL/DSCR respectively). Taken together, the image swap perturbs the model’s answer more than a no-image-dependency baseline would predict, but the resulting change does not reliably reflect the new image’s actual content — a pattern more consistent with noise-like sensitivity to the input changing than with substantive extraction of the substituted patient’s findings.

KAB framework metrics. On v3, answer recoverability (Section 3.1) is near ceiling for every model on every phase (96-100%), meaning a model’s stated reasoning text almost always predicts its own answer; as already noted in Section 8, this establishes internal answer-reasoning consistency, not that the reasoning is image-derived. Ontology term coverage is bimodal across phases rather than uniformly informative: 15-48% on AIA and LIL (open descriptive imaging language, only partially covered by RadLex/NCIt vocabulary) versus 94-98% on DSCR, PJRF, and TCM (whose correct answers are drawn from a small set of fixed clinical phrases, e.g. RANO cate-

gories and Stupp-protocol terminology, that the ontologies cover almost completely). This split reflects each phase’s answer-vocabulary structure more than a difference in reasoning quality, consistent with the term-validity-only caveat already stated in Section 8. Answer-term overlap is low and variable (0-73% across models and phases), reflecting its definition as overlap with the correct answer’s own terminology rather than a broader measure of explanation quality. Feedback-adaptation was triggered rarely in this run and only on LIL (1-4 of 52 patients per model, 0 triggers on every other phase), too sparse to support a conclusion either way about self-correction. Gated chain completion is not yet reported: every v3 run completed to date used gating disabled (nogate) so that ungated per-phase accuracy could be compared cleanly against the text-only ablation; a gated v3 run, needed to report upstream-term repetition and chain-completion rate, has not yet been executed and is left for future work.

Two findings that sharpen this reading. Two additional v3 results bear on how the numbers above should be read, in different directions. First, a trivial majority-class baseline (always answering *progressive disease*) scores 67% on DSCR, because LUMIERE’s glioblastoma cohort is overwhelmingly progressing and this is both the correct answer and typically the longest option; all four models sit at or below this baseline (Gemma-3-27B 65%, Llama-4-Scout 58%, Gemma-3-12B 56%, MedGemma-4B 14%). This is a construct-validity caveat on the DSCR numbers specifically: no model demonstrates RANO-assessment skill above what the class imbalance alone would predict. It should not be read as further evidence for or against image grounding. Second, LIL accuracy is at or near the four-option chance rate of 25% for three of four models (Gemma-3-12B 23%, Gemma-3-27B 27%, Llama-4-Scout 25%; MedGemma-4B 40%), and text-only LIL accuracy is statistically the same (23-38%). Unlike the DSCR finding, this result speaks directly to grounding: models are not merely indifferent to the image on LIL, they are unable to extract the lesion volume-change direction from it even when the task asks for exactly that comparison.

Reading against the pre-registration. Section 7 pre-registered that text-only accuracy equivalent to image-present accuracy within a pre-specified margin, evidence classified mostly as copied rather

than independently supported, and a low, non-truth-tracking counterfactual flip rate would together indicate that correct answers on this item set are not meaningfully image-grounded. All four checks are now complete on v3. The first two conditions are met as stated: every model’s image-minus-text-only accuracy difference falls within a ± 8 pp patient-clustered margin (not within a tighter ± 5 pp margin), and independently-supported evidence accounts for only 0-8% of correct answers’ stated claims. The third is only partly met as originally phrased: LIL/DSCR flip rates (6-42% across models) are not uniformly *low* in absolute terms, exceeding the AIA no-image-dependency baseline (10-17%) for three of four models. What the flip-rate check does establish cleanly is *non-truth-tracking*: the pooled rate at which a flipped answer matches the substituted patient’s actual facts (35.6%, 95% CI [25.6, 47.1]) sits entirely below chance, not above it. We read the overall pattern as consistent with the non-grounding hypothesis on balance, on the strength of the third check’s non-truth-tracking result specifically rather than its flip-rate magnitude, but flag the flip-rate mismatch with the original pre-registered wording explicitly rather than silently treating a partial match as a full one. This reading covers the open-weight model suite; results for the proprietary models (Section 6) have not yet been run.

Limitations

This paper reports the KAB framework, the chain-validity problem it surfaces in an existing benchmark, the dataset construction method used to address it, the image-grounding test suite of Section 7, and results from all four of that suite’s checks (Section 8) on the open-weight model suite; it does not yet report these results for the proprietary models described in Section 6. The leak-controlled LUMIERE item set (v3, Section 5) that these results are drawn from was itself a correction to an earlier, leakier draft (v2), exactly the class of construct-validity problem that motivated the grounding-test suite of Section 7 in the first place, and a reason to continue reading any result on this item set as provisional pending further validation. Separately, the counterfactual-image substitution’s flip-rate magnitude did not cleanly match the pre-registered “low” criterion (Section 8); we read the check’s non-truth-tracking result as the decisive part of that check rather than treating the mismatch as satisfied by

assertion.

Several additional limitations apply to the framework and dataset construction as currently specified. First, ontology term coverage (Section 3.1) is a string- and n-gram-matching metric against a fixed vocabulary, not a semantic one, and it only scores the fraction of a model’s *recognized* terms that are valid ontology entries; unrecognized assertions are excluded from its denominator entirely, so a response can score highly while containing substantial unsupported content, and the metric will separately undercount a model that expresses a correct clinical concept in phrasing not present in either ontology. Neither this metric nor answer-term overlap establishes that the underlying clinical claim is correct for the patient, only that the vocabulary used is valid and, respectively, on-target; we expect this to be a point of reviewer scrutiny. Second, in OmniBrainBench, PJRF and TCM are a small fraction of the overall 6,823-question dataset (64 and 51 questions respectively, well under 5% of cases each), so results reported on that dataset’s PJRF/TCM subsets carry low statistical power and should be reported with confidence intervals rather than pooled with the larger phases, which would mask exactly the degradation signal the framework is designed to detect; the LUMIERE chain dataset does not have this imbalance, since every patient contributes exactly one PJRF and one TCM item alongside the other three phases, but its overall $n = 52$ still means per-phase confidence intervals are wide, and we report them rather than a single pooled number for the same reason. Third, the tumor segmentations underlying LUMIERE’s lesion-location and volume facts are produced by automated tools (DeepBraTumIA, HDGLIO-AUTO) and are not manually verified per-timepoint, whereas the RANO treatment-response ratings are expert-assigned; we state this distinction explicitly rather than implying uniform annotation quality across fact types. Fourth, LUMIERE is a single-center cohort, and a benchmark built on it inherits whatever institution-specific imaging protocol and patient population characteristics that entails. Fifth, the answer-recoverability probe (Section 3.1) is, by construction, only able to detect an inconsistency between a model’s stated reasoning and its answer; it cannot on its own establish that consistent reasoning is image-derived, nor that it is causally responsible for the answer at all, which is why Section 7 tests grounding through ablation, evidence auditing, and counterfactual substitution

rather than through this probe alone. Sixth, the evidence fact-check’s *copied* category is a deterministic text-overlap match against the stem, options, and chain context (Section 7); a model that inspects the image, identifies a real finding, and expresses it using wording that overlaps the answer options is still classified as copied, so this category does not cleanly separate genuine visual verification from restating what was shown, and the audit’s category counts have not been validated against independent expert annotation. Finally, the model suite (Section 6) does not include any open-weight model requiring multi-GPU tensor parallelism, so this is not a constraint on the results reported here; a larger open-weight model would reopen that question.

Ethics Statement

This work evaluates the diagnostic reasoning of general-purpose and medical-domain language models on real, de-identified patient imaging data; it does not deploy, recommend, or endorse any model for clinical use, and no results in this paper should be interpreted as a clinical validation of any evaluated system. LUMIERE is used under its published non-commercial research license, and all patient data is accessed in its already de-identified, publicly released form; no new identifiable patient information is collected.

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A A Worked Patient Example

To make the abstract description of items, chain context, and evidence auditing in Sections 3-8 concrete, this appendix walks through one complete case end to end: Patient-002’s leak-controlled (v3) five-phase chain, as answered by Gemma-3-27B (the best-performing open-weight model), together with the evidence fact-check’s actual (audit-tool-generated, not hand-picked) classification of the model’s stated reasoning at each phase.

The patient. Patient-002 is a 71-year-old female with glioblastoma (MGMT unmethylated, IDH status unavailable) who underwent complete resection of the enhancing tumor followed by Stupp-protocol chemoradiation. The post-operative re-baselined LIL comparison is week-040-2 (baseline) versus week-047 (follow-up): total lesion volume increases from 27,937mm³ to 57,570mm³ (+106%), with the enhancing core and necrotic components moving from the Superior Parietal Lobule at baseline to the Left Cerebral White Matter at follow-up. The week-047 RANO rating is progressive disease. Recorded overall survival from surgery is 48 weeks.

AIA. *Question:* which imaging modalities were available. *Correct/model answer:* T1, contrast-enhanced T1, T2, and FLAIR (correct). The model’s stated visual_grounding for this item, unprompted, also volunteers a lesion location: “a mass lesion in the right temporal lobe.” No part of the AIA question or options mentions location, so this claim is neither asked for nor copied from anything shown to the model. Checked against the same structured facts used to draft the item, it is wrong on two counts: the true baseline location is the Superior Parietal Lobule/Left Cerebral White Matter (not temporal), and the true laterality is left, not right. Running the evidence fact-check’s hemisphere check on this claim in isolation (Section 7) labels it *laterality_flip*: a real, audit-caught contradiction, reported under that label rather than as ordinary fabrication per the paper’s own convention (Section 7), since a display-orientation ambiguity cannot be ruled out for this claim in the same way it can for later ones below.

LIL. *Question:* how did the lesion’s location and volume change between baseline and follow-up. *Correct/model answer:* shifted predominantly to the Left Cerebral White Matter, +106% (correct). The model’s reasoning restates the same fabricated

baseline location from AIA (“the initial lesion is located in the right temporal lobe”), now unchanged across a full phase boundary, and concludes with the correct answer’s own location and percentage. Run through the fact-check, every checkable claim in this response, the repeated “right temporal lobe” and the “106%” figure alike, is labeled *copied*: not because the audit missed the AIA-level contradiction, but because by LIL, both claims are now present in the model’s own prior-turn output or the shown answer options, which is exactly what the audit is designed to detect and exactly why it cannot, by itself, distinguish a genuinely re-verified observation from one simply carried forward.

DSCR. *Question:* RANO response category at week-047. *Correct/model answer:* progressive disease (correct). The model’s evidence again repeats the “right temporal lobe” framing and the radiology-report measurement it was shown, and the fact-check again labels every claim *copied*. The final answer is correct, and consistent with the (still-wrong) location claim the model has now repeated across three consecutive phases without correction.

PJRF. *Question:* expected overall survival from surgery. *Correct answer:* approximately 12 months (48-52 weeks) — age over 70 as the dominant adverse factor despite complete resection. *Model answer:* approximately 9 months (36-40 weeks) — the option attributing a *particularly* poor prognosis to the same unmethylated-MGMT/age-over-70 combination. This is the one incorrect answer in the chain: both options describe the same adverse risk factors in the same direction, and the model picked the more pessimistic of two plausible-sounding buckets rather than the one matching the recorded 48-week survival exactly.

TCM. *Question:* appropriate management at this time. *Correct/model answer:* initiate second-line therapy given evidence of true progression (correct).

Reading this case. Taken as a whole, this single chain illustrates the gap Section 8 reports in aggregate: 4 of 5 final answers are correct, but the evidentiary trail behind the LIL and DSCR answers rests on a spatial claim introduced at AIA that the fact-check itself flags as contradicting the ground truth, not on independently re-derived observation at each phase. We did not select this patient for this property after the

1406 fact; it is Patient-002, the first case in the evalua-
1407 tion order, and its fact-check labels above are ex-
1408 actly what tools/audit_reasoning_facts.py
1409 outputs when run on it, not a paraphrase.